

Video Game Sales Analysis

A Data-Driven Exploration Using Visualization

DS612 — Interactive Data Visualization

Group 3

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1. Introduction

What determines whether a video game becomes a global success—or fades into obscurity?

Every year, thousands of video games are released across multiple platforms and regions. While some rise to achieve massive commercial success and cultural recognition, many others disappear without leaving a significant impact. This raises a fundamental question: **is success in the gaming industry driven by identifiable patterns, or is it inherently unpredictable?**

This study begins with a simple assumption: that factors such as **platform choice, genre, and regional demand** influence a game's performance. However, instead of accepting these assumptions at face value, we examine them through a structured, data-driven investigation.

The journey unfolds across four analytical phases. It begins with an exploration of **platform-level dynamics**, followed by an analysis of **regional differences in consumer preferences**. The focus then shifts to **Japan as a distinct market**, before concluding with an examination of **industry-level dynamics**, including publisher strategies and revenue concentration.

What emerges is not a fixed formula for success, but a broader pattern—one shaped by interaction, alignment, and context. **Game success is not the result of a single defining factor, but a combination of multiple interconnected influences.**

This report is not merely an analysis of sales data. It is an exploration into how the structure of the gaming industry shapes outcomes—why certain games succeed, why others fail, and what these patterns reveal about an evolving global market.

Narrative Flow: This report follows a structured analytical progression:

- **(1)** Analysing platform-level dynamics,
- **(2)** Examining regional differences in sales,
- **(3)** Investigating Japan as a distinct market, and
- **(4)** Exploring industry-level concentration and publisher strategies.

2. Context & Background

With the rapid growth of the global video game industry, thousands of games are released across multiple platforms each year. Despite this expansion, only a small fraction of titles achieve significant commercial success, while the majority struggle to gain visibility in an increasingly competitive market.

Understanding what drives success in this environment has become an important question for developers, publishers, and platform providers. Factors such as platform choice, genre, and regional demand are often assumed to influence performance. However, it remains unclear whether these factors alone can explain why certain games succeed while others fail.

One of the defining characteristics of the gaming industry is the presence of strong **regional differences**. At the same time, the industry is influenced by a relatively small number of **dominant publishers** who control a large share of total sales. Their strategies, including regional focus and portfolio management, further impact how games perform across markets.

Given these complexities, it becomes necessary to move beyond simple assumptions and examine how these factors interact. **This study aims to provide a structured understanding of the underlying patterns that define success in the global video game market.**

3. Problem Statement

While several factors are believed to influence the success of video games, there is no clear evidence that any single attribute can consistently predict a game's performance.

Common assumptions suggest that elements such as **platform choice, genre, and regional demand** play a significant role in determining success. However, it remains unclear whether these factors alone are sufficient to explain why certain games achieve high global sales while others fail to gain traction.

This study aims to investigate whether these commonly assumed factors truly influence performance, and to explore deeper structural patterns that define success in the global video game industry.

4. Objectives

The primary objective of this study is to develop a structured understanding of the factors that influence success in the global video game market.

- Analyse the relationship between **platforms, genres, and regional sales** and their impact on global performance
- Examine commonly held assumptions regarding what drives the success of video games
- Identify patterns that go beyond individual factors by exploring interactions across multiple dimensions
- Understand how **regional behavior and publisher strategies** contribute to market outcomes
- Derive a broader structural understanding of how success is distributed within the industry

Visualization Goals:

This study employs a combination of **single-frame** to uncover patterns in the dataset.

Single-variable plots are used to analyse distributions and identify general trends, while multi-variable visualizations—such as correlation plots and interaction-based charts—are used to capture relationships between multiple factors. These visual tools play a critical role in transforming raw data into meaningful insights.

5. Dataset Description

The dataset used in this study consists of video game sales records across multiple platforms, genres, and regions. Each entry represents a video game along with its corresponding sales performance across major markets, providing a comprehensive view of the global gaming industry.

The dataset used in this project is the VGChartz Global Video Game Sales dataset, containing information about video game titles, their platforms, genres, publishers, and regional sales.

Variable	Data Type	Description	Notes
Rank	Integer	Sales rank (1 = best-selling)	Continuous 1–16,600
Name	String	Full game title	16,598 unique entries
Platform	Categorical	Gaming console or PC platform	31 unique values
Year	Float/Int	Year of release	271 missing values (1.6%)
Genre	Categorical	Game genre classification	12 genres
Publisher	String	Publishing company name	579 unique; 58 missing
NA_Sales	Float	North America sales in millions	Range: 0 – 41.49 M
EU_Sales	Float	Europe sales in millions	Range: 0 – 29.02 M
JP_Sales	Float	Japan sales in millions	Range: 0 – 10.22 M
Other_Sales	Float	Rest-of-world sales in millions	Range: 0 – 10.57 M
Global_Sales	Float	Total worldwide sales in millions	Range: 0.01 – 82.74 M

Figure 1: Description of Dataset Variables including data types, meanings, and key observations

Target Variable

- **Global Sales:** Total worldwide sales of a game, representing its overall success.

Derived Features

To enhance analytical depth and support hypothesis-driven analysis, additional features were constructed:

- **Regional Share:** Proportion of sales contributed by each region:

$$\text{JP Share} = \frac{\text{SUM(JP Sales)}}{\text{SUM(Global Sales)}}, \quad \text{NA Share} = \frac{\text{SUM(NA Sales)}}{\text{SUM(Global Sales)}}, \quad \text{EU Share} = \frac{\text{SUM(EU Sales)}}{\text{SUM(Global Sales)}}$$

- **Over-Indexing Metric:** Measures regional preference differences:

$$\text{Difference} = \frac{\text{SUM}(\text{JP Sales}) - \text{SUM}(\text{NA Sales})}{\text{SUM}(\text{Global Sales})}$$

- **Favour:** This variable categorizes whether a game or genre performs better in Japan or North America based on relative sales differences:
 - JP Favored: If Difference > 0
 - NA Favored: If Difference < 0

Publisher Type: Classification of publishers into:

- Japanese Publishers (e.g., Nintendo, Capcom, Square Enix)
- Foreign Publishers

Aggregated Metrics: Average sales, total sales, and counts computed across platforms, genres, and publishers to support comparative analysis.

These derived features enable a more comprehensive analysis by capturing relationships that are not directly visible from the raw data. The dataset, being multi-dimensional and continuous, is well-suited for both statistical analysis and visualization-driven exploration.

6. Data Preprocessing

Before conducting the analysis, the dataset was examined and prepared to ensure **consistency, accuracy, and reliability** of results. This step is essential to eliminate inconsistencies that may affect the analysis.

Handling Missing Values

```
: df.isnull().sum()

: Rank      0
  Name      0
  Platform  0
  Year    271
  Genre     0
  Publisher 58
  NA_Sales  0
  EU_Sales  0
  JP_Sales  0
  Other_Sales 0
  Global_Sales 0
  dtype: int64
```

Figure 2: Missing Values in Dataset

The dataset contains a small number of missing values, primarily in the *Year* and *Publisher* attributes. Given the relatively small proportion, these records were handled appropriately to maintain data integrity.

Steps Performed

Data Type Verification: All variables were checked for correct data types and consistency.

Validation of Feature Ranges: Sales values were verified to ensure logical consistency across regions and global totals.

These preprocessing steps ensure that the dataset is clean, structured, and ready for meaningful analysis.

7. Exploratory Data Analysis (EDA)

This section explores the dataset to identify patterns, distributions, and relationships. The objective is to develop an intuitive understanding of the data before moving into hypothesis-driven analysis.

7.1 Statistical Overview



```
df.describe()
```

	Rank	Year	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
count	16598.000000	16327.000000	16598.000000	16598.000000	16598.000000	16598.000000	16598.000000
mean	8300.605254	2006.406443	0.264667	0.146652	0.077782	0.048063	0.537441
std	4791.853933	5.828981	0.816683	0.505351	0.309291	0.188588	1.555028
min	1.000000	1980.000000	0.000000	0.000000	0.000000	0.000000	0.010000
25%	4151.250000	2003.000000	0.000000	0.000000	0.000000	0.000000	0.060000
50%	8300.500000	2007.000000	0.080000	0.020000	0.000000	0.010000	0.170000
75%	12449.750000	2010.000000	0.240000	0.110000	0.040000	0.040000	0.470000
max	16600.000000	2020.000000	41.490000	29.020000	10.220000	10.570000	82.740000

Figure 3: Descriptive Statistics of Numerical Features

The statistical summary highlights the distribution of key numerical variables. It shows that while most games have low sales values, a few titles achieve exceptionally high performance.

Insight: The large gap between median and maximum values indicates strong skewness in the dataset.

7.2 Correlation Analysis

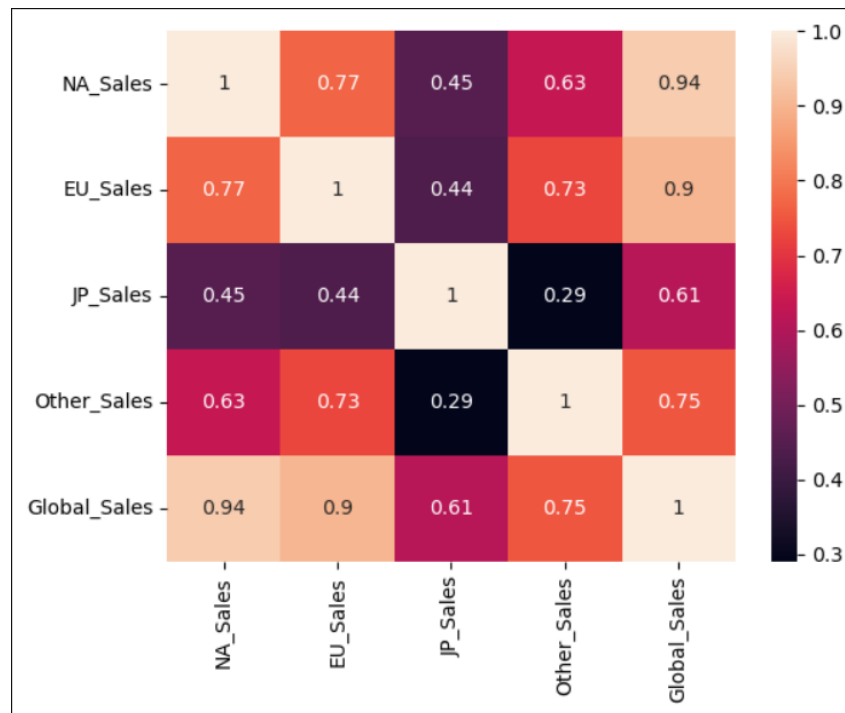


Figure 4: Correlation Between Sales Across Regions

The correlation analysis shows:

- Strong relationship between **NA Sales** and **Global Sales**
- Strong relationship between **EU Sales** and **Global Sales**
- Moderate relationship for **JP Sales**

Insight: Global success is largely driven by North American and European markets, while Japan exhibits distinct behavior.

7.3 Key Observations

- Regional markets contribute differently to overall success
- No single factor explains performance on its own

Transition: While these insights reveal general patterns, they do not fully explain the underlying drivers of success. This motivates a shift toward a **hypothesis-driven analysis** to explore deeper relationships.

8. Hypothesis 1: Platform dominance shapes sales outcomes

Objective: To examine whether platform choice, temporal trends, and genre-platform relationships influence the global sales performance of video games.

Q1: Does platform choice affect a game's global sales ceiling?

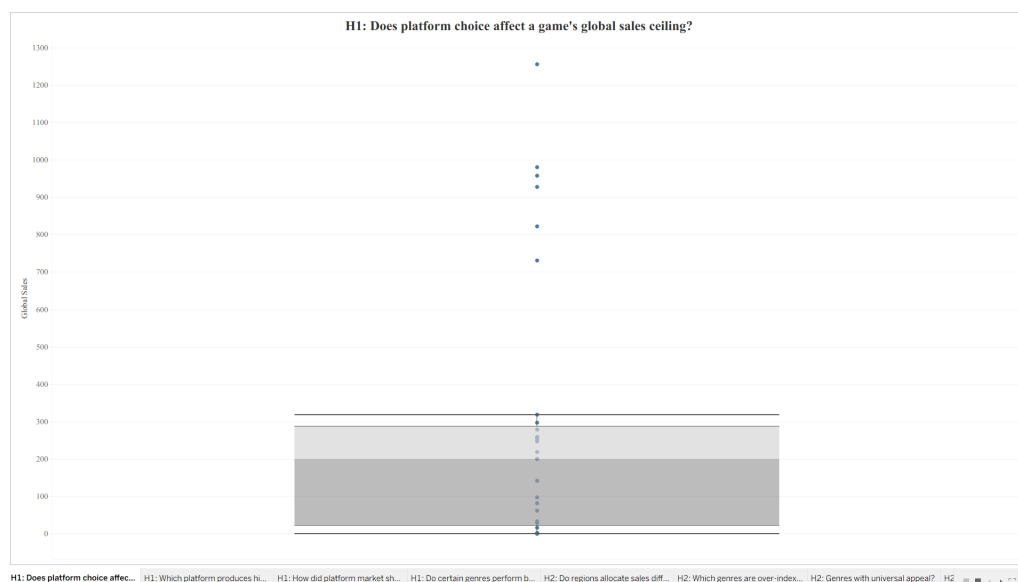


Figure 5: Platform vs Global Sales Distribution

We begin with the most fundamental question — whether platform choice limits the maximum success a game can achieve.

Platform does not impose a strict upper bound on success; blockbuster performance is rare but platform-agnostic. Platforms with higher upper whiskers/outliers = higher ceiling Example: “PS2 and X360 show higher extreme values → stronger sales potential”

Insight: The distribution of global sales shows that most games cluster within a moderate range, while a small number of titles emerge as extreme outliers with exceptionally high sales.

Q2: Which platform produces highest average sales per game?

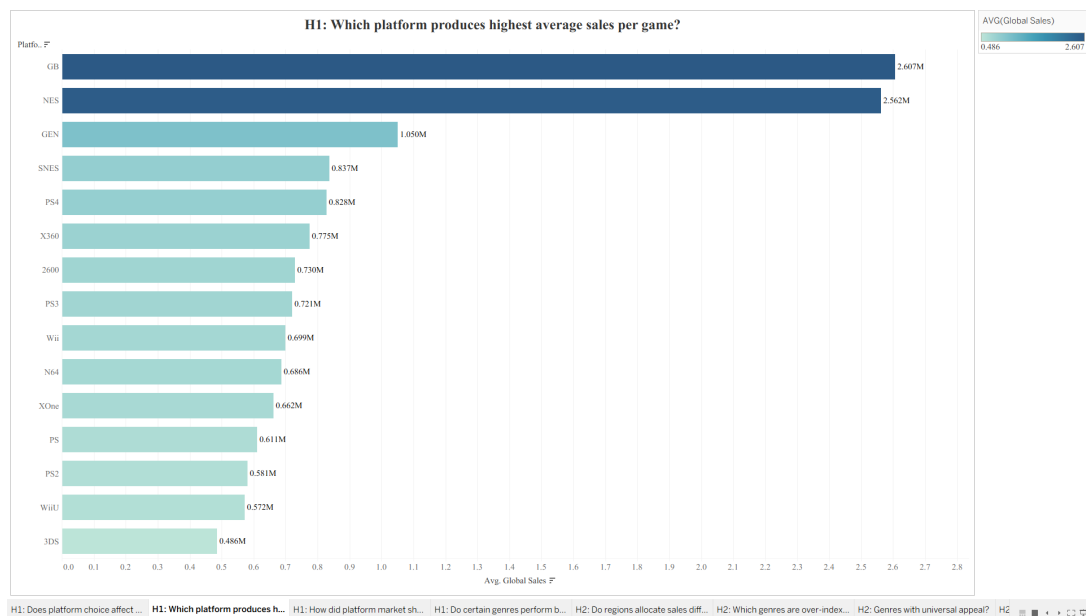


Figure 6: Average Global Sales per Platform

Shifting from extremes to averages, we observe that legacy platforms such as **GB** and **NES** show the highest average global sales. Example: “Nintendo platforms show higher average due to strong exclusives”

Insight: Average global sales are sensitive to outliers and catalog size, making them a limited measure of consistent success. High average = efficient platform (not just volume-driven)

Q3: How did platform market share shift over time?

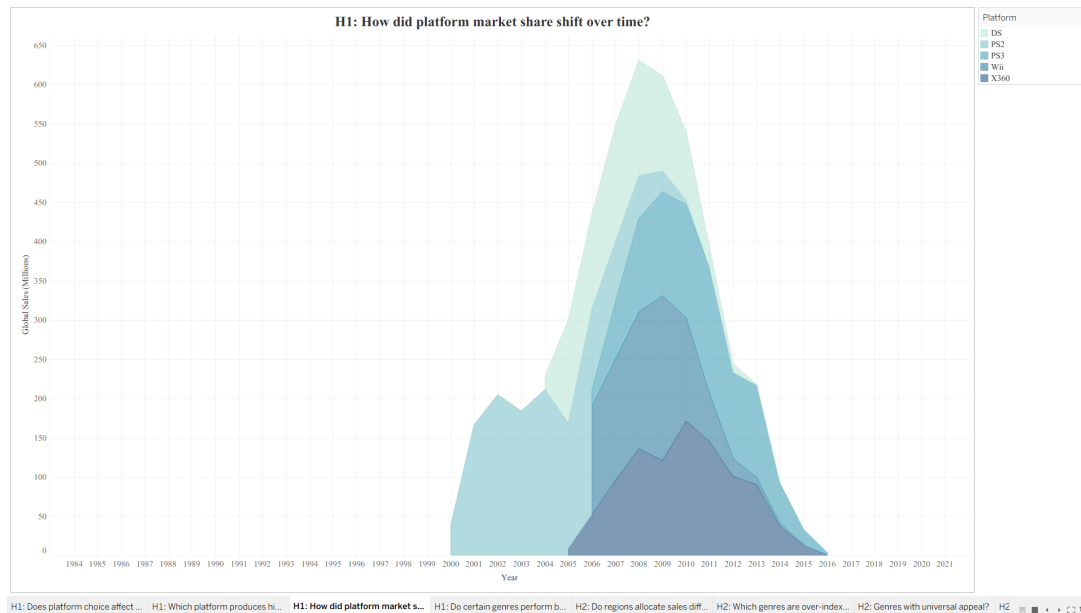


Figure 7: Platform Market Share Over Time

Earlier periods are dominated by platforms such as PS2 and DS, while later years show the emergence of PS3, Xbox 360, and Wii.

A distinct peak in total sales is observed around 2008–2010, followed by a noticeable decline.

Insight: Platform success is highly dependent on lifecycle stages and technological evolution. The temporal analysis highlights clear generational transitions in platform dominance

Q4: Do certain genres perform better on specific platforms?

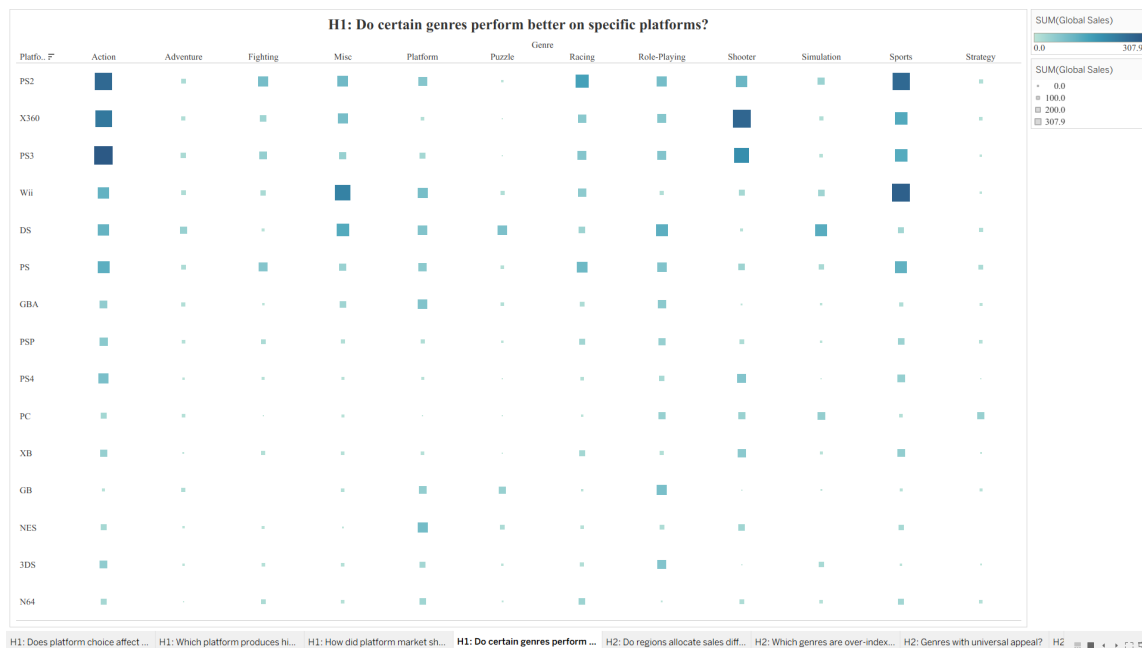


Figure 8: Genre Performance Across Platforms

Bright spots = strong combinations Example insights: “Sports games dominate on PlayStation platforms” “Nintendo platforms overperform in Platform Puzzle genres”

Insight: Success is not driven by genre or platform alone, but by their interaction. The analysis reveals clear interaction effects between genre and platform. Certain genres such as action and sports dominate across multiple platforms, while others remain niche.

Conclusion of the Hypothesis:

Platform selection significantly impacts both the sales ceiling and average performance of games. Legacy platforms like PS2 achieved high peak sales, while newer platforms redistributed market share over time. Additionally, genre-platform alignment plays a critical role, with certain genres consistently outperforming on specific platforms.

This reinforces a critical insight: no single dimension — platform, time, or genre — can independently explain global sales performance. Instead, success emerges from a layered interaction of multiple elements.

This transition motivates a deeper investigation into regional dynamics and market behavior in the next hypothesis.

9. Hypothesis 2: Regional markets have divergent genre preferences

Objective: To investigate whether game success is driven by balanced global appeal or strong regional preferences across genres and time.

Q1: Do regions allocate sales differently across genres?

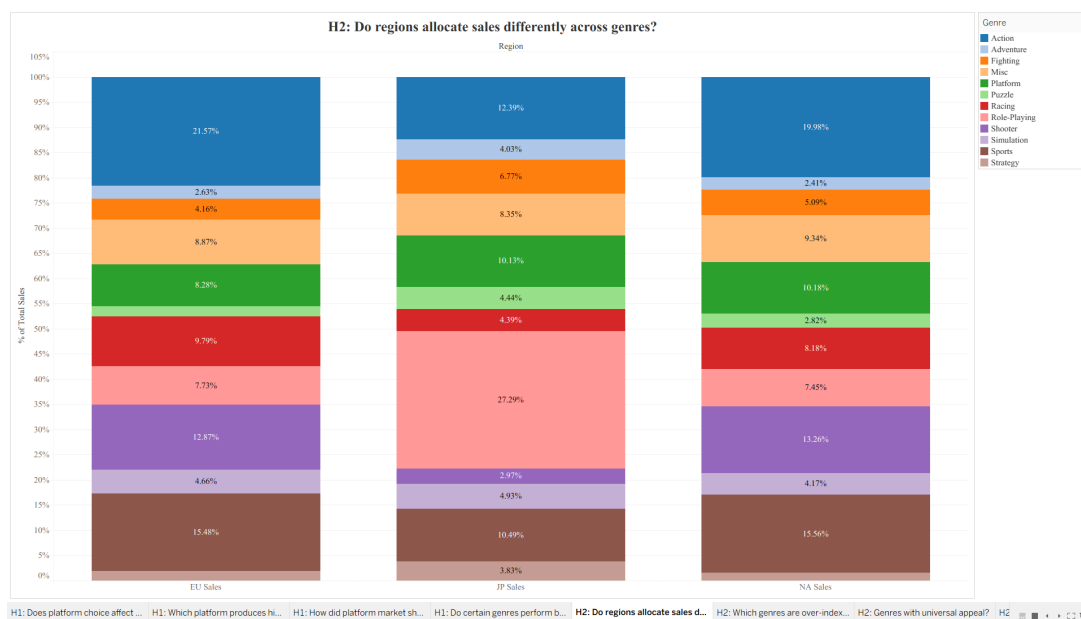


Figure 9: Regional Sales Distribution Across Genres

The distribution of sales across genres reveals clear regional differences. North America and Europe show strong dominance in genres such as **Action**, **Sports**, and **Shooter**, indicating a preference for fast-paced and competitive gameplay.

In contrast, Japan displays a significantly different pattern, with a strong emphasis on **Role-Playing** games and relatively lower contributions in shooter-based genres.

Insight: Regional markets are not homogeneous; each region exhibits distinct genre preferences.

Q2: Does divergence vary by platform or era?

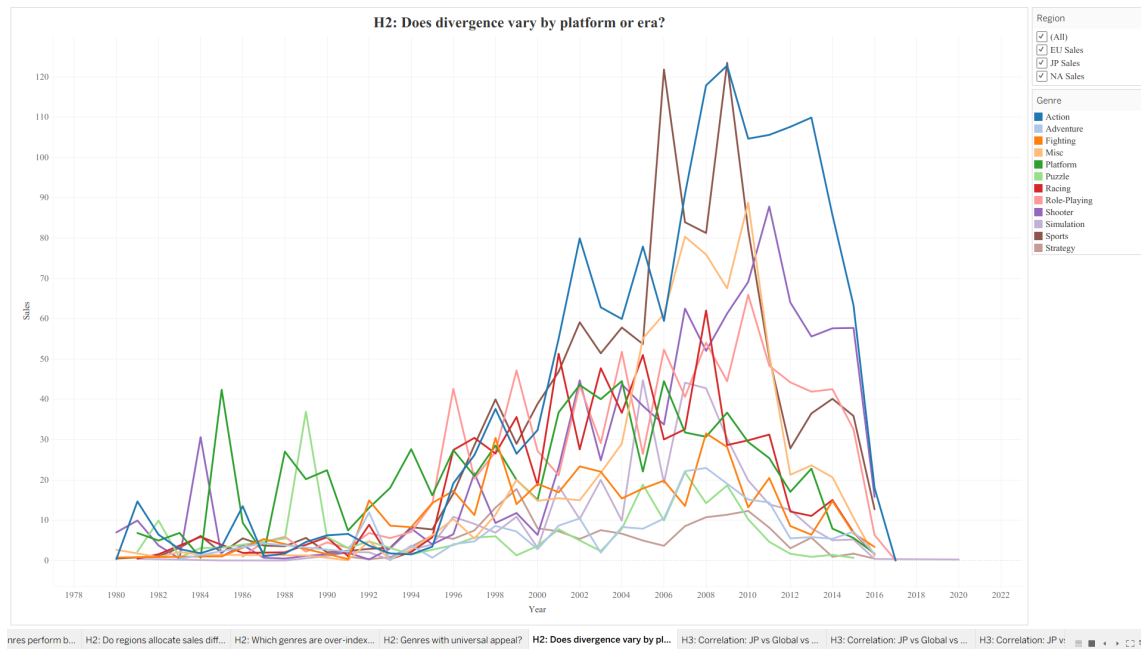


Figure 10: Genre Trends and Regional Divergence Over Time

The temporal analysis reveals that regional divergence is not constant, but evolves across different phases of the gaming industry. During peak periods, particularly around 2005–2010, multiple genres show simultaneous growth across regions. However, the magnitude of this growth differs significantly, with North America and Europe expanding more aggressively compared to Japan.

While some genres display synchronized trends across regions, others exhibit strong regional spikes, indicating localized preferences. Additionally, shifts in dominant platforms across eras further influence these patterns, as different platforms cater to different audiences and markets.

Insight: Regional divergence is dynamic and shaped by both temporal evolution and platform transitions, rather than remaining fixed over time.

Q3: Which genres are over-indexed in Japan vs NA?

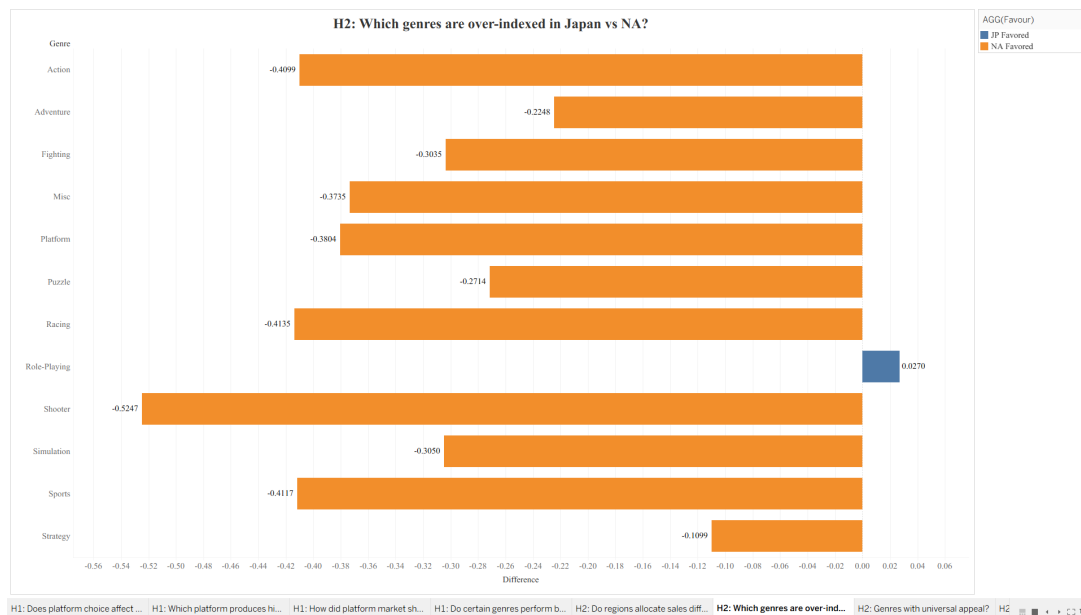


Figure 11: Genre Over-Indexing: Japan vs North America

The over-indexing analysis compares regional preferences by measuring the relative difference in genre performance between Japan and North America. Positive values indicate a stronger preference in Japan, while negative values reflect a stronger preference in North America. The magnitude of the value represents the strength of this preference difference.

The results reveal a clear regional divide. Genres such as **Shooter**, **Sports**, and **Action** are strongly over-indexed in North America, indicating a preference for fast-paced and competitive gameplay. In contrast, **Role-Playing** games show a strong positive bias toward Japan, reflecting the region's historical inclination toward narrative-driven and strategic experiences.

Additionally, genres with values close to zero exhibit relatively balanced performance across both regions, suggesting limited regional bias and a more universal appeal.

Insight: Genre preferences are highly region-specific, with only a few genres achieving balanced performance across markets.

Q4: Which genres show universal appeal?

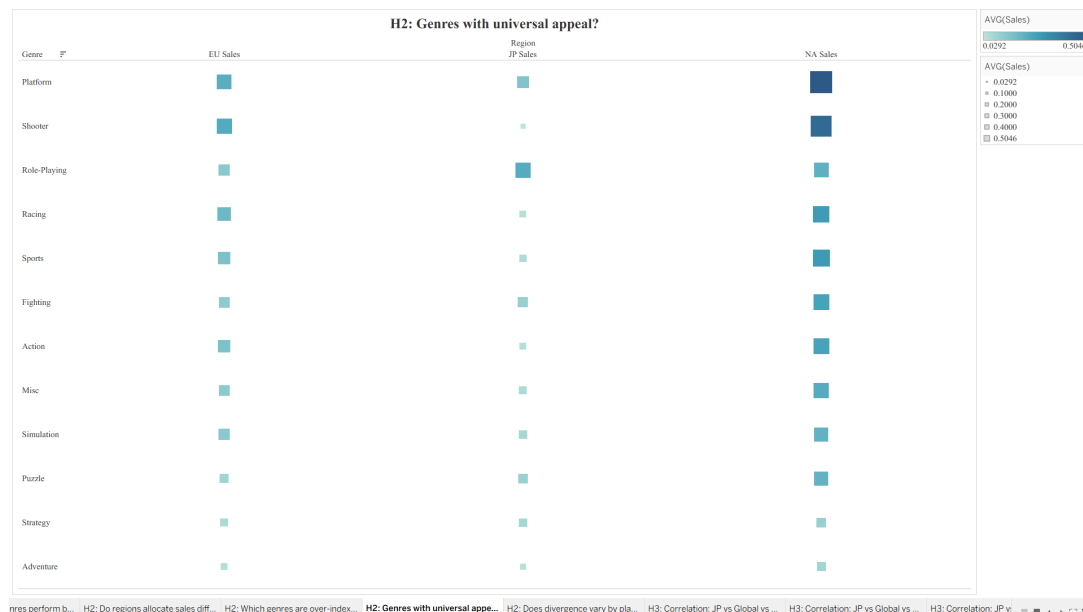


Figure 12: Genres with Cross-Regional Appeal

The analysis of average sales across regions reveals that only a limited number of genres maintain relatively balanced performance globally. Example: “Platform games demonstrate universal appeal with consistently high sales across NA, EU, and JP.”

While most genres show strong regional bias, a few — such as **Action** and **Misc** — exhibit moderate consistency across regions, though still not perfectly balanced.

Insight: Truly universal genres are rare; most success is region-specific. Some genres are consistently strong across all regions.

Conclusion of the Hypothesis:

The analysis reveals that game success is not driven by universal appeal alone, but by how well a game aligns with regional preferences.

Different regions prioritize different genres, and these preferences evolve over time. While a few genres achieve moderate global consistency, most exhibit strong regional bias, particularly between Japan and North America.

This challenges the idea of a single global formula for success and highlights the importance of strategic regional targeting.

These findings mark a transition from balance-based reasoning to deeper structural analysis, leading to the next hypothesis focused on correlations and publisher-level

strategies.

10. Hypothesis 3: Japan is an outlier market that behaves independently

Objective: To analyse how regional success, particularly in Japan, translates to global performance and how publisher origin and time trends influence market dynamics.

Q1: Correlation — JP vs Global vs NA/EU

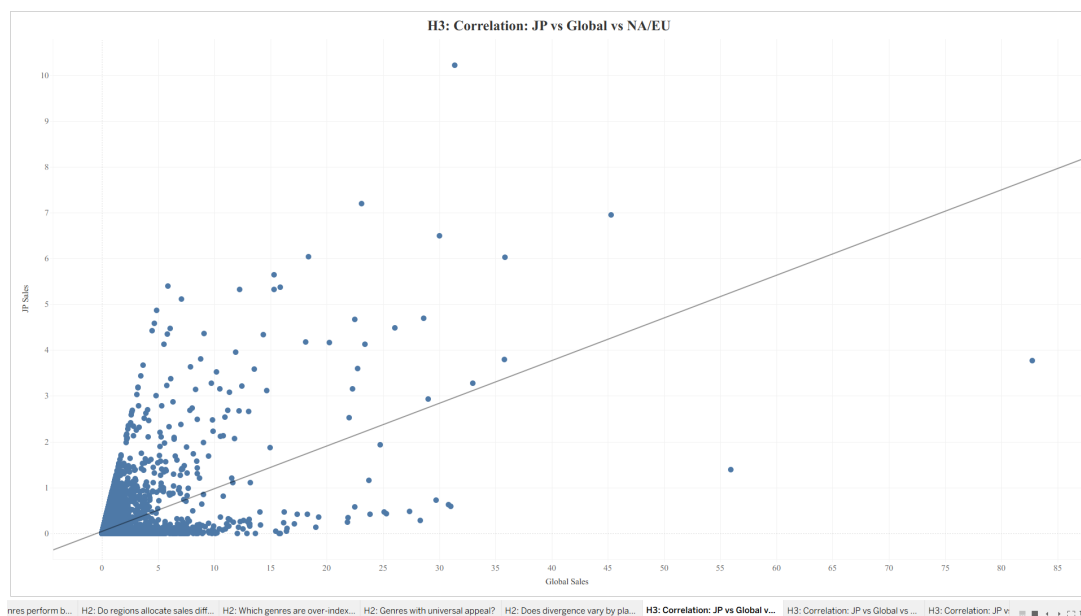


Figure 13: JP Sales vs Global Sales

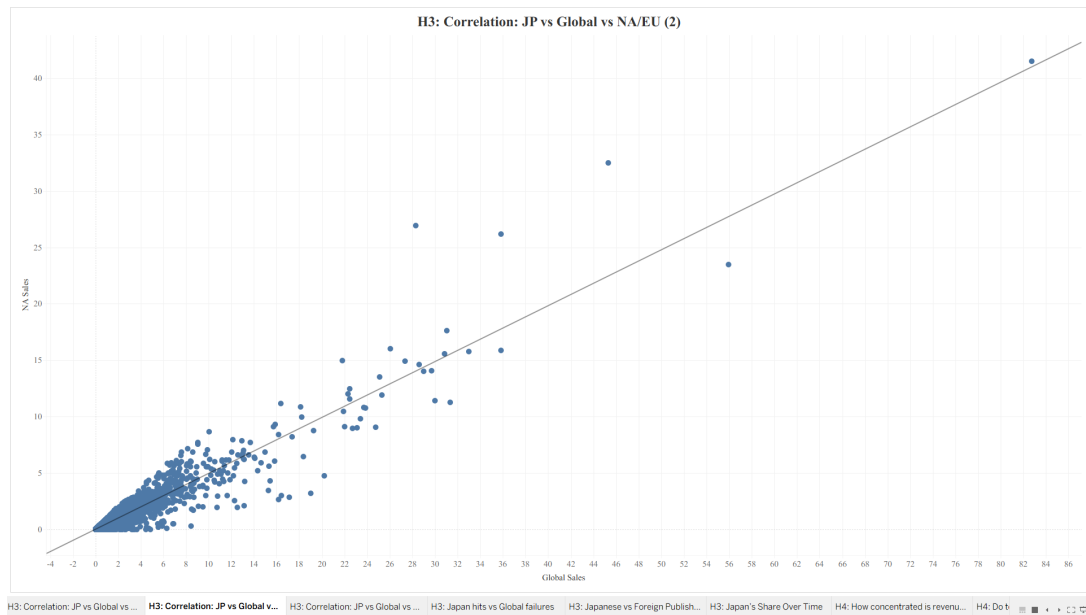


Figure 14: NA Sales vs Global Sales

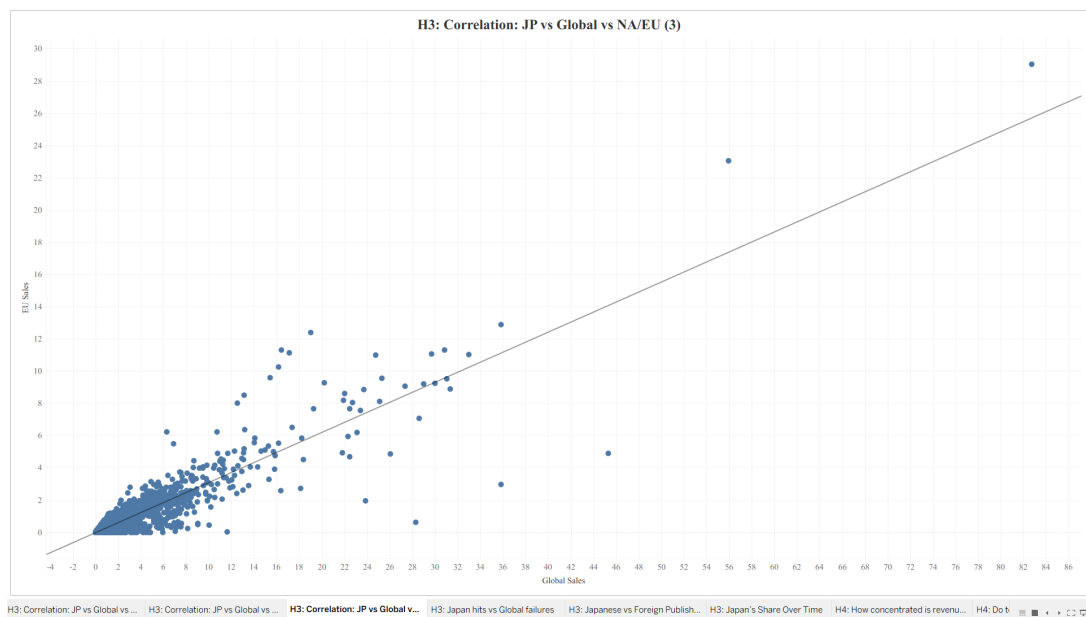


Figure 15: EU Sales vs Global Sales

The correlation analysis reveals a clear asymmetry between regions. While North America and Europe show a strong linear relationship with global sales, Japan exhibits a much weaker and scattered pattern.

Many games achieve moderate success in Japan without corresponding global impact, whereas high global sales are almost always accompanied by strong NA/EU performance.

Insight: Western markets act as primary drivers of global success, while Japan operates as a more independent ecosystem.

Q2: Japan Hits vs Global Failures

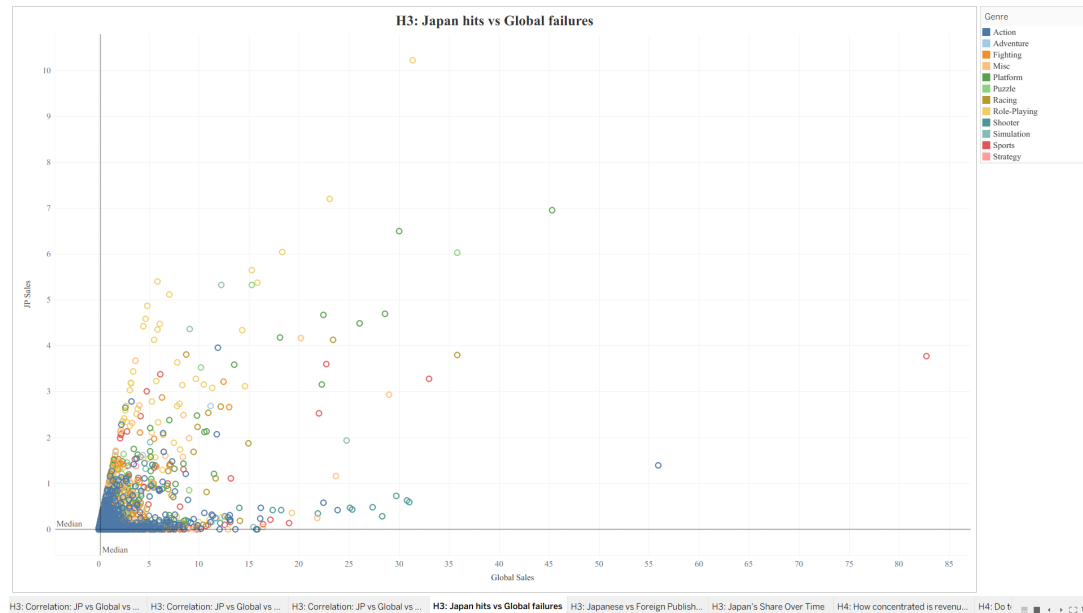


Figure 16: Japan Hits vs Global Performance

The scatter plot comparing Japanese sales and global sales provides a quadrant-based view of market performance across regions. Each quadrant represents a distinct pattern of success.

Games located in the **top-right quadrant** (high JP, high global) represent globally successful titles that perform well across all markets. In contrast, the **top-left quadrant** (high JP, low global) highlights games that achieve strong success in Japan but fail to scale globally. These represent region-specific hits unique to the Japanese market.

On the other hand, the **bottom-right quadrant** (low JP, high global) captures titles that perform well globally but have limited success in Japan, indicating Western-dominated hits. The **bottom-left quadrant** represents low-performing games across all regions.

A key observation is the presence of multiple titles in the top-left quadrant, confirming that certain games resonate strongly with Japanese audiences while failing to achieve broader international appeal.

Furthermore, the distribution of data indicates that a significant number of games record negligible or zero sales in Japan. This reinforces the idea that Japan is a highly

selective and distinct market with limited cross-regional penetration.

Insight: Japan exhibits unique success patterns that are not strongly aligned with global trends.

Q3: Japanese vs Foreign Publishers

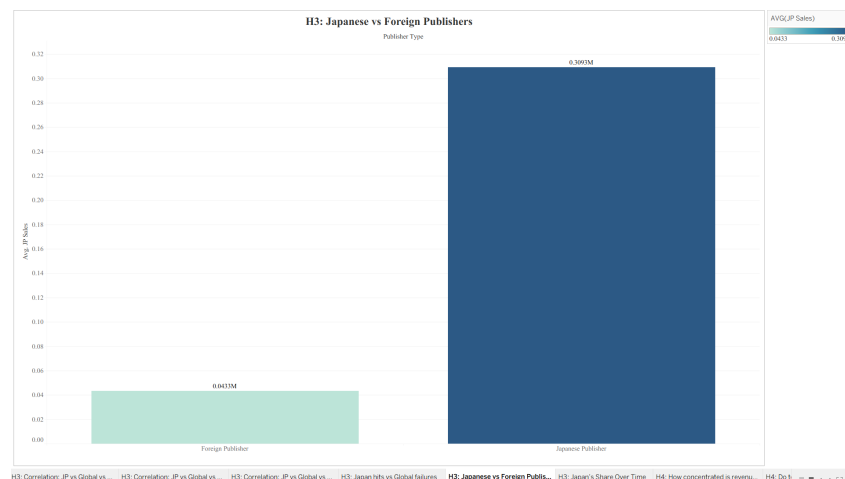


Figure 17: Average JP Sales by Publisher Type

The comparison between Japanese and foreign publishers reveals a significant difference in regional performance.

Japanese publishers show substantially higher average sales in Japan, indicating strong domestic dominance. In contrast, foreign publishers exhibit lower JP sales, suggesting limited penetration into the Japanese market.

Insight: Publisher origin strongly influences regional success.

Q4: Japan's Share Over Time

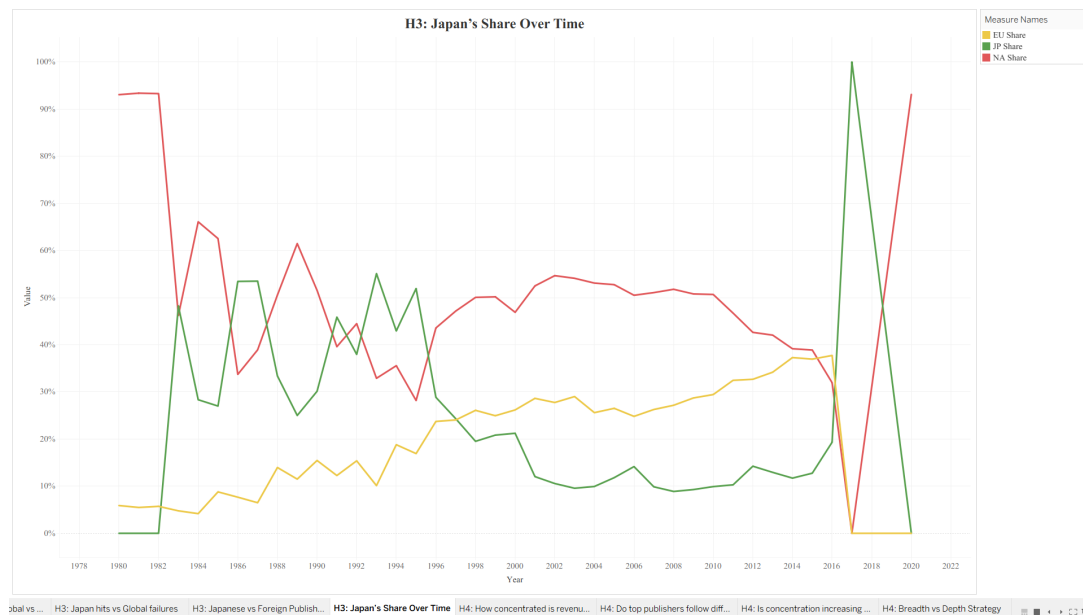


Figure 18: Regional Share Trends Over Time

This chart compares the regional share of global sales over time. Japan's declining trend relative to North America and Europe suggests a shift in global market dominance.

The rise of globalized gaming ecosystems and international distribution reduces the dominance of region-specific markets.

Insight: The global market has shifted toward Western dominance over time.

Conclusion of the Hypothesis:

The analysis supports the hypothesis that Japan operates as a partially independent gaming market. Correlation analysis reveals weaker alignment between Japanese and global sales compared to North America and Europe. Additionally, certain genres and titles perform disproportionately well in Japan, while Japanese publishers maintain stronger domestic performance. Over time, Japan's share of global revenue has declined, further highlighting its distinct market trajectory.

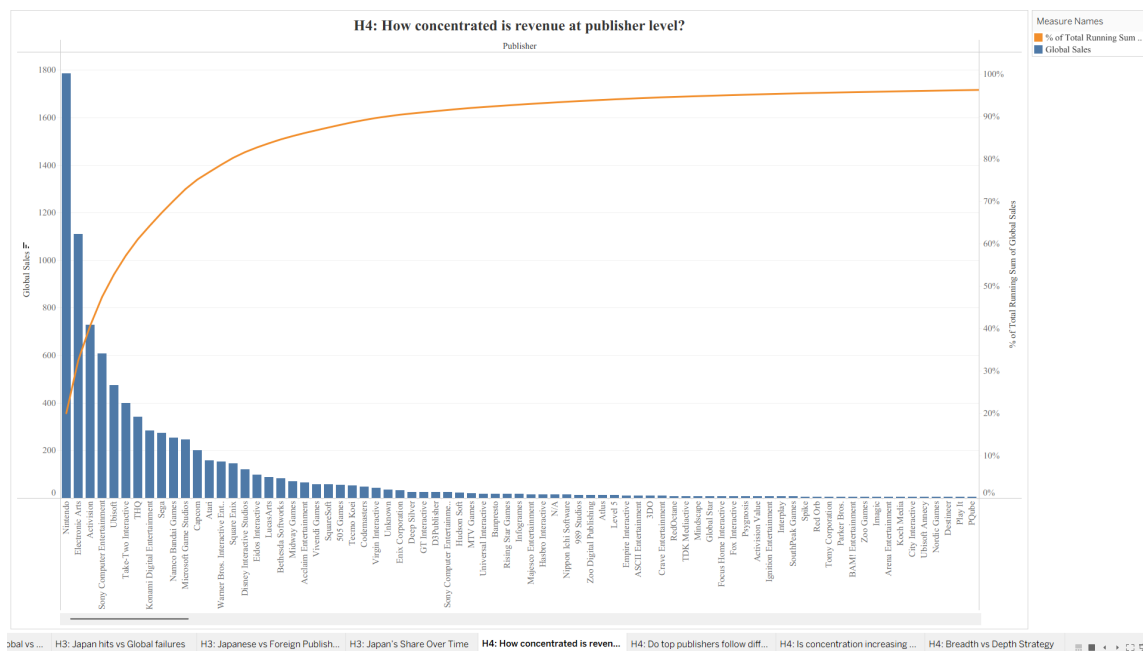
Together, these findings reveal that global success is not merely a function of product quality, but of regional alignment, market access, and strategic positioning.

This naturally leads to the final hypothesis, which investigates whether the industry ultimately operates under a winner-takes-all structure dominated by a few key players.

11. Hypothesis 4: The Industry Follows a Winner-Takes-All Structure

Objective: To analyse whether a small number of publishers dominate the gaming industry and to understand the strategic patterns behind this dominance.

Q1: How concentrated is revenue at publisher level?



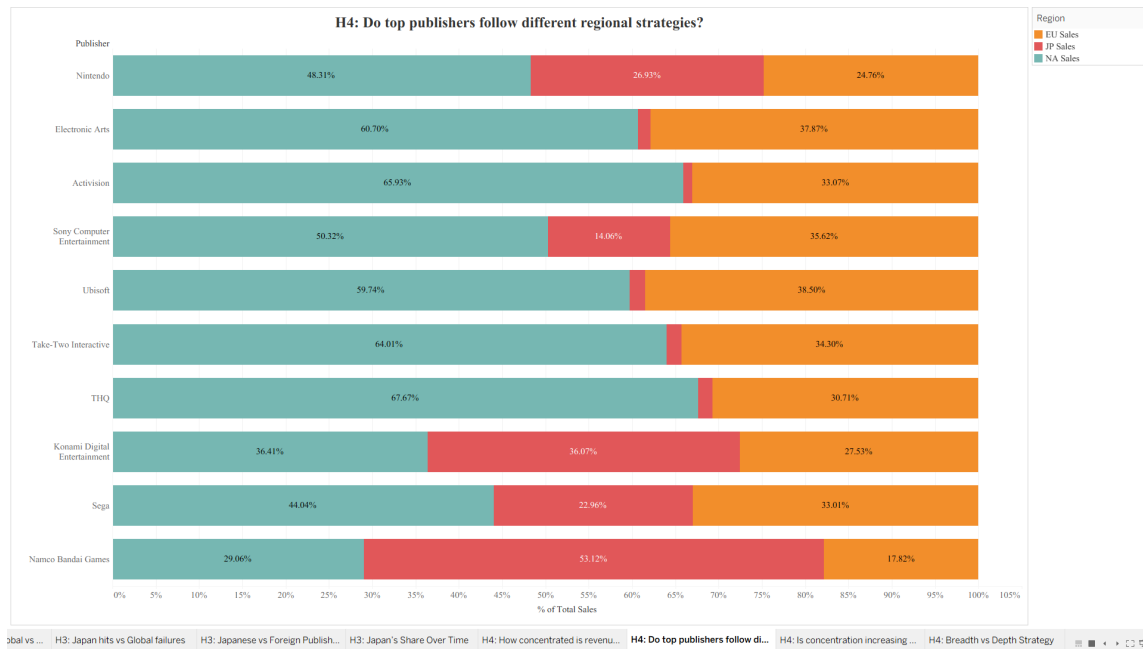


Figure 20: Regional Sales Distribution by Publisher

The regional distribution of sales highlights clear differences in publisher strategies. Some publishers demonstrate strong dominance in specific regions, while others maintain a more balanced global presence.

Namco Bandai Games shows strong dependence on the Japanese market, indicating domestic dominance. Electronic Arts focuses heavily on Western markets with limited presence in Japan. Some publishers adopt a globally diversified strategy across regions.

Insight: Publishers adopt distinct regional strategies based on market strengths.

Q3: Is concentration increasing over time?

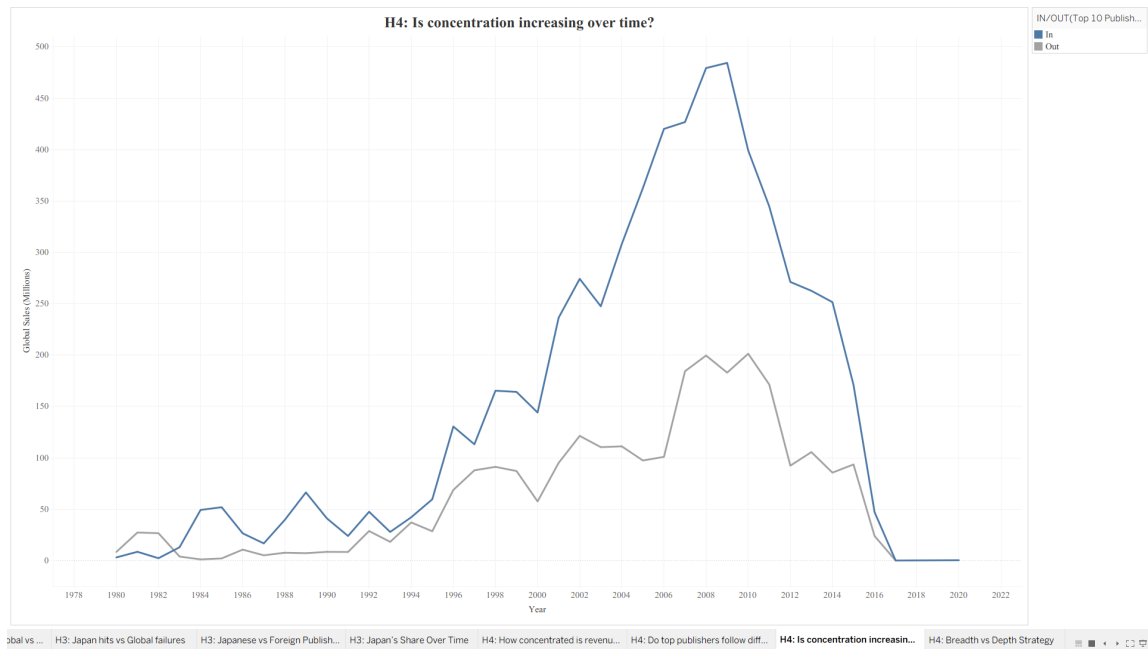


Figure 21: Top Publishers vs Others Over Time

The temporal trend shows how sales contributions from top publishers compare to the rest of the market over time. The gap between the two indicates the level of concentration.

Increasing → more monopoly Decreasing → more competition

During peak years, particularly around the late 2000s, the dominance of top publishers becomes more pronounced. However, the trend also shows periods of decline, suggesting phases of increased competition.

Insight: Market concentration fluctuates over time but repeatedly returns to dominance by top publishers.

Q4: Breadth vs Depth Strategy

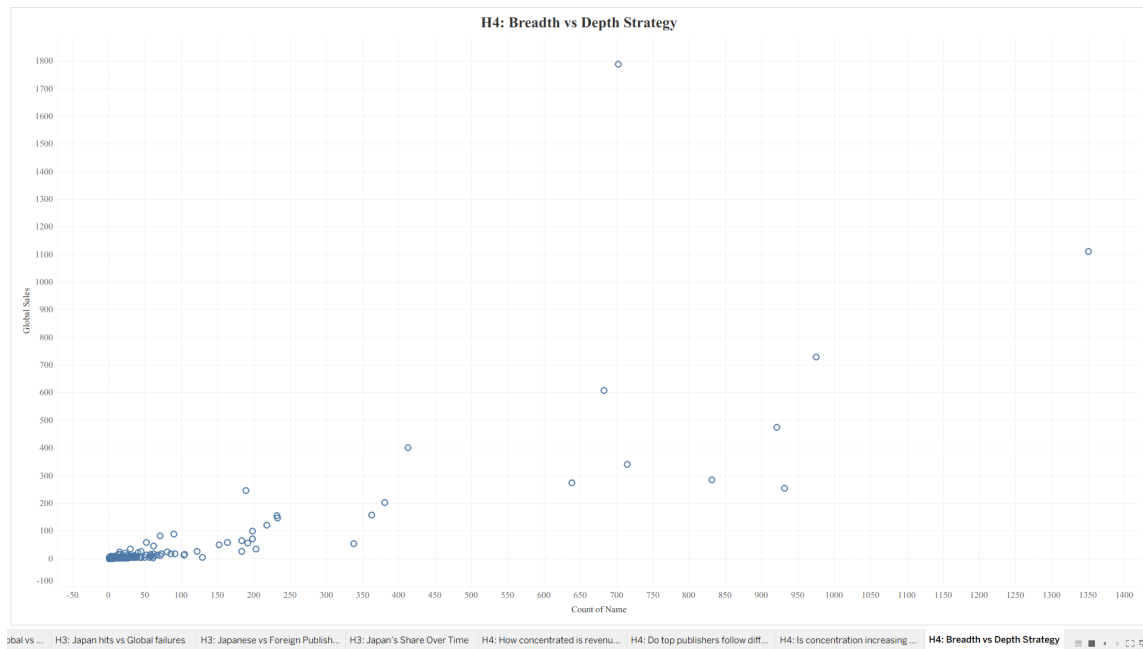


Figure 22: Number of Games vs Global Sales

The relationship between the number of games released and total sales reveals different strategic approaches adopted by publishers.

Some publishers achieve high sales with a large portfolio of games, representing a volume-driven strategy. Others generate high revenue from a smaller number of titles, indicating a blockbuster-focused approach.

Position Meaning: Top-right Many games + high sales → dominant Top-left Few games + high sales → blockbuster strategy Bottom-right Many games + low sales → volume strategy

Insight: Success is achieved through different strategies, but dominance often requires either scale or high-impact releases.

Conclusion of the Hypothesis:

The analysis reveals a highly concentrated industry structure, where a small number of publishers capture a disproportionate share of global revenue. These leading publishers employ varied strategies, with some focusing on regional dominance and others achieving global reach. While some rely on a large portfolio of titles, others succeed through blockbuster releases. Trends over time suggest evolving levels of market concentration, reflecting shifts in competitive dynamics.

The gaming industry is highly concentrated, with a small group of publishers driving the majority of global sales, confirming the presence of a winner-takes-all ecosystem.

12. Conclusion: Understanding the Structure of Success in the Gaming Industry

This analysis shows that success in the video game industry cannot be explained by a single factor. Instead, it results from the interaction of platform choice, regional preferences, genre alignment, and publisher strategy.

Individual factors such as platform or genre provide only partial explanations. Strong performance typically emerges when multiple elements align effectively, rather than from any single variable.

Regional differences play a crucial role. North America and Europe largely drive global success, while Japan represents a distinct and selective market with unique preferences. This highlights that success is not universal, but region-dependent.

At the industry level, a clear **winner-takes-all structure** is observed, where a small number of publishers dominate global revenue through varied strategies such as regional focus, global expansion, and portfolio management.